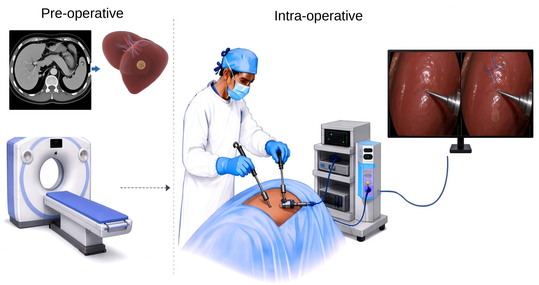
Vision Intelligence for Image Guided Surgery

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Motivation



Pre-operative CT or MRI (left) and image-guided minimally invasive surgery (right).

- Minimally invasive surgery uses small incisions, a camera, and specialized tools, so patients recover faster.
- The surgeon sees only the organ surface on a screen, with a limited view and no touch feedback.
- Image guidance brings the 3D plan from CT or MRI into the live camera view.

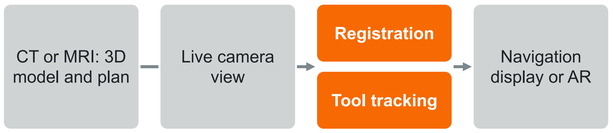
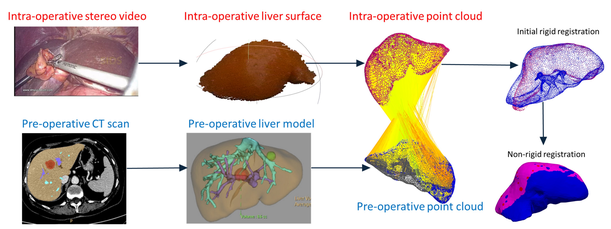


Image-guided surgery workflow. This work addresses the two orange steps.

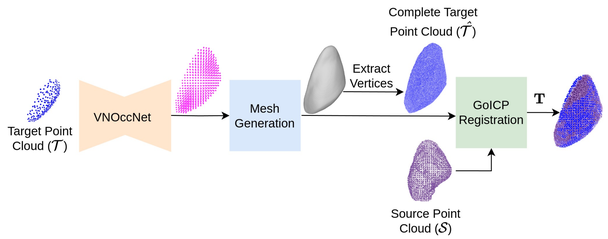
Problem: a partial view

- Tumors and vessels lie beneath the liver surface.
- Registration finds the transform that overlays the pre-operative model on the camera view.
- The camera sees only part of the liver, so a partial surface must be matched to a complete model.
- Point-matching methods (3DMatch, Lepard, Predator) also need an even point density.



Overview of the 3D-3D registration process. Intra-operative stereo video and pre-operative CT each give a liver point cloud. Initial rigid registration brings them closer; non-rigid registration then accounts for tissue deformation.

Method



The partial intra-operative point cloud is completed by VN-OccNet, turned into a mesh, and aligned to the pre-operative surface with GO-ICP.

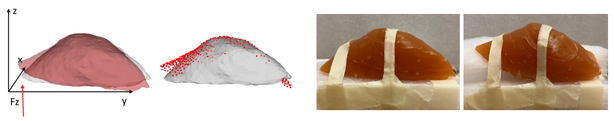
1. **Complete:** VN-OccNet predicts the missing liver surface from the partial points.
2. **Mesh:** MISE and Marching Cubes give a watertight surface with even point density.
3. **Register:** GO-ICP rigidly aligns the completed surface with the pre-operative model.

Why VN-OccNet?

- Rotation equivariant: the camera viewpoint and patient position change, so the partial view can arrive in any orientation.
- Built-in mesh generation avoids uneven point density.

Data

- **In silico phantoms** train and validate VN-OccNet.
- **In vitro phantoms** test registration.



In silico liver phantom

In vitro liver phantoms

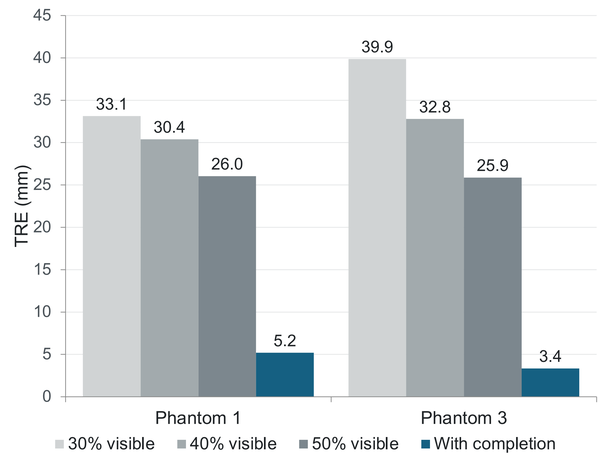
Surface completion results

Metric	OccNet	VN-OccNet
F-Score (higher is better)	0.07 ± 0.07	**0.51 ± 0.18**
Chamfer dist. (lower is better)	29.43 ± 13.41	**3.25 ± 1.83**
IoU (higher is better)	0.35 ± 0.18	**0.87 ± 0.06**

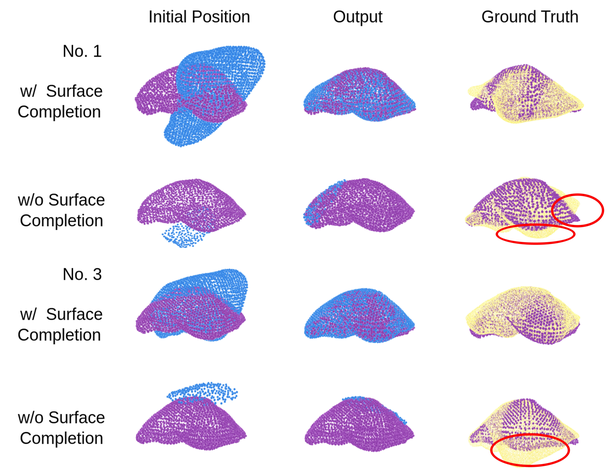
Trained with Z-axis rotations, tested with arbitrary SO(3) rotations not seen in training.

- VN-OccNet keeps its accuracy on unseen orientations; OccNet does not.

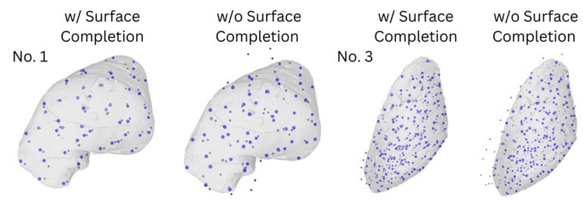
Results: registration



Completion cuts TRE from 26 to 40 mm (30 to 50% visible) to 3.4 to 5.2 mm. Target registration error (TRE, mm, lower is better) between target and registered source fiducial markers. Grey bars: partial surface only.



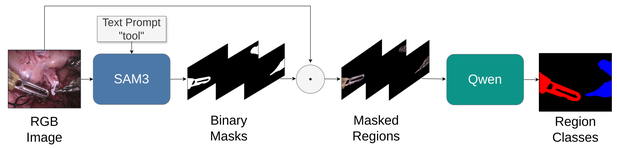
Initial position, registration output, and ground truth overlay for phantoms No. 1 and No. 3, with and without surface completion. Source (purple), target (blue). Red circles mark misalignment without completion.



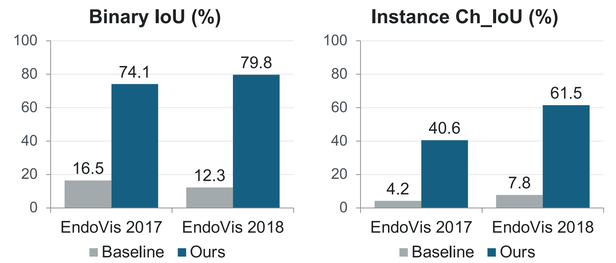
Fiducial markers on phantoms 1 and 3: target (blue) and source (purple) markers sit close together only with surface completion.

Surgical tool segmentation

- Image-based tracking estimates the tool's 6 degree-of-freedom pose from the camera, with no extra sensors.
- It first needs a mask of the tool pixels. Hand-labeled masks are costly and hard to keep consistent under occlusion, blood, and lighting change.



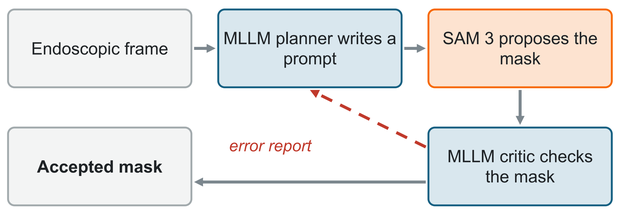
SAM3 with the text prompt "tool" finds tool masks. Qwen, a vision-language model fine-tuned on these masked regions, names each instrument. No mask annotation is used.



Baseline: SAM3 prompted with instrument names. Ours: SAM3 "tool" (binary) plus Qwen (instance). On EndoVis 2017, our binary IoU (74.1) is close to a fully supervised UNet (75.4). Main errors: false positives, merged neighboring tools, fragmented masks, and missed tools.

Conclusion and next steps

- Completing the liver surface before registration handles partial views and improves TRE by 6×, with no manual interaction.
- A text prompt plus a fine-tuned vision-language model gives instrument masks without mask labels.
- **Next:** agentic mask generation, where a model proposes, checks, and revises each mask.



Planned agentic loop. A mask still rejected after N rounds is flagged for a human.

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Acknowledgments Supported by NIH NIGMS Award No. R35GM128877 and NSF CBET Award No. 2245152. Computing resources from RIT Research Computing. Thanks to Zixin Yang and Kelly Merrell, co-authors on the registration work.